

Softmax#

<https://pytorch.org/docs/stable/generated/torch.nn.modules.activation.Softmax>

Description:

Applies the Softmax function to an n-dimensional input Tensor. Rescales them so that the elements of the n-dimensional output Tensor lie in the range [0,1] and sum to 1. Softmax is defined as: When the input Tensor is a sparse tensor then the unspecified values are treated as -inf. Input: (*) (*) where * means, any number of additional dimensions

Function Signature

```
class torch.nn.modules.activation.Softmax(dim=None)[source]#
```

Shape:

Returns

Parameters

Return type

Returns:

Tensor

Code Examples

```
>>> m = nn.Softmax(dim=1)
>>> input = torch.randn(2, 3)
>>> output = m(input)
```

Notes & Warnings

NOTE This module doesn't work directly with NLLLoss, which expects the Log to be computed between the Softmax and itself. Use LogSoftmax instead (it's faster and has better numerical properties).

Detailed Documentation

Documentation

Applies the Softmax function to an n-dimensional input Tensor.

Rescales them so that the elements of the n-dimensional output Tensor lie in the range [0,1] and sum to 1.

Softmax is defined as:

When the input Tensor is a sparse tensor then the unspecified values are treated as -inf.

Input: $(*)^{(*)} (*)$ where $*$ means, any number of additional dimensions

Output: $(*)^{(*)} (*)$, same shape as the input

a Tensor of the same dimension and shape as the input with values in the range [0, 1]

dim (int) – A dimension along which Softmax will be computed (so every slice along dim will sum to 1).

This module doesn't work directly with NLLLoss, which expects the Log to be computed between the Softmax and itself. Use LogSoftmax instead (it's faster and has better numerical properties).

Return the extra representation of the module.

Runs the forward pass.

